Preprocessing Briefing - Multi-Person UWB Localization

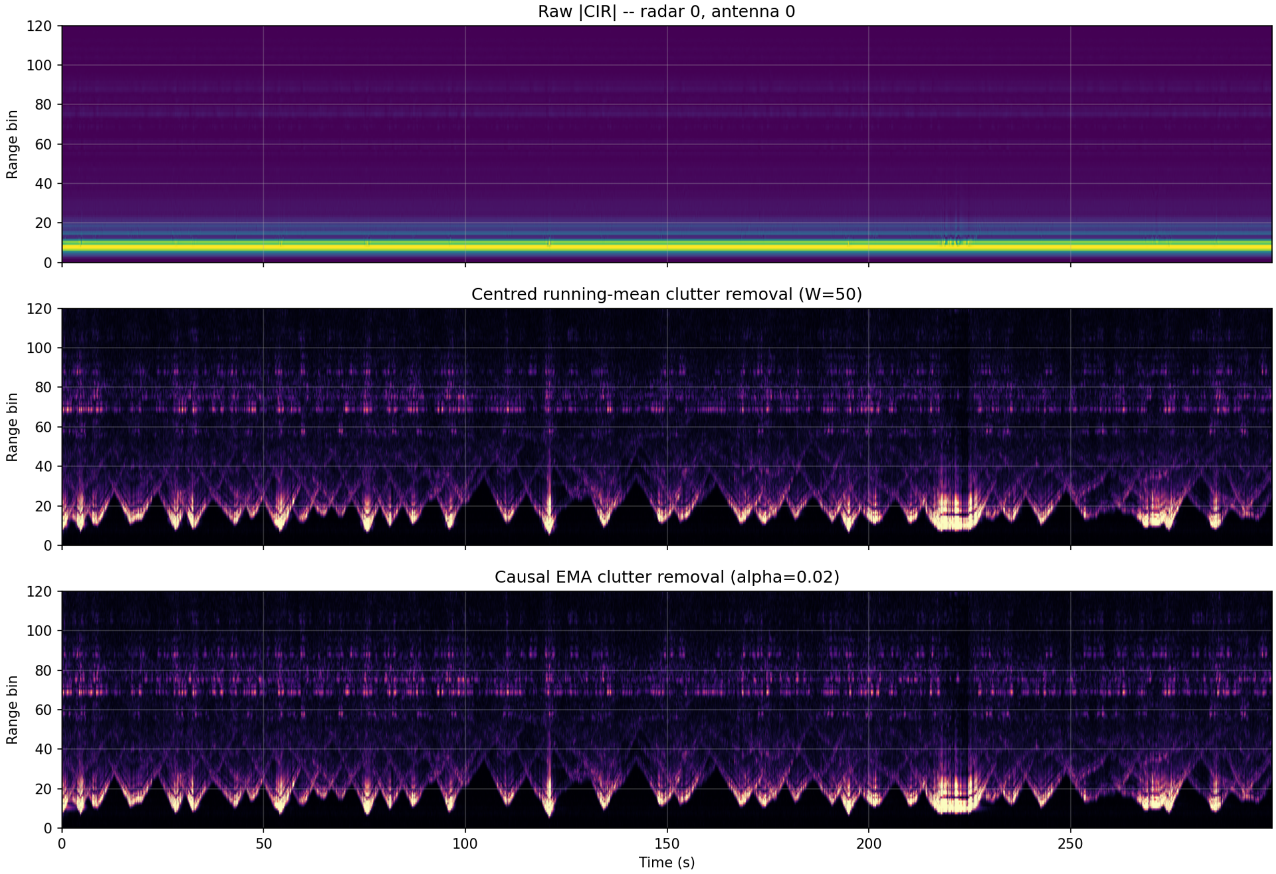
Generated 2026-04-30 12:11 - Pipeline = causal EMA clutter removal -> range-bin crop -> z-score - Train: 16 windows, Val: 4 windows, Test: 4 windows

1. Pipeline parameters

Stage	Method	Parameters
Clutter removal	Causal EMA	alpha = 0.02 (tau ~ 50 frames = 2 s); warmup = 50 frames
Range-bin crop	Energy-driven	keep [6, 111) -> 105 bins (99% energy)
Normalisation	z-score	per (radar, antenna, channel); fit on train split only

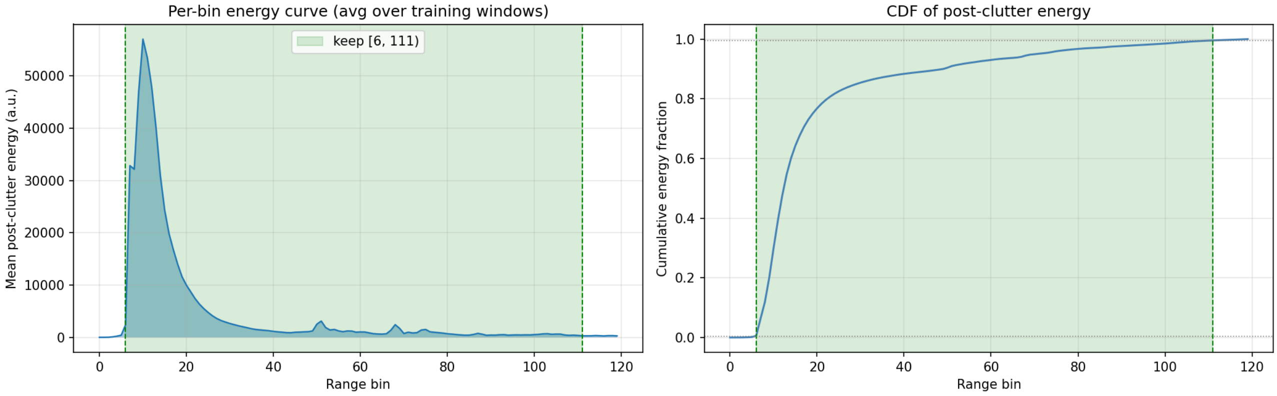
2. Clutter removal: running-mean (offline) vs EMA (causal)

The training-time and inference-time pipelines must be identical, so we adopt the causal EMA. The figure below confirms it produces a comparable post-clutter image to the offline centred running mean used in the EDA.



3. Range-bin energy curve (post-EMA)

Per-bin |I/Q|^2 energy averaged across the 16 training windows, computed after causal EMA clutter removal. The chosen crop [6, 111) preserves 99.34% of the energy and reduces input range from 120 to 105 bins (87.5% of original).



Region	Bins	Energy fraction
Near-field tail (dropped)	[0, 6)	0.15 %
Kept	[6, 111)	99.34 %
Far-field tail (dropped)	[111, 120)	0.51 %

4. Normalisation statistics (z-score)

Per (radar, antenna, channel) mean and std of the post-clutter, post-crop signal, computed on the training split only. Stats stored in `normalization_stats.npz`.

R	A	mean I	std I	mean Q	std Q
0	0	0.00	93.44	0.00	94.29
0	1	0.00	36.73	0.00	36.89
0	2	-0.00	13.31	-0.00	13.46
1	0	0.00	82.45	0.00	80.52
1	1	0.00	26.44	0.00	26.59
1	2	-0.00	11.62	-0.00	11.91
2	0	0.00	97.93	-0.00	101.20
2	1	0.00	28.44	0.00	30.63
2	2	-0.00	12.60	-0.00	12.39
3	0	-0.00	109.34	-0.00	109.54
3	1	0.00	29.09	0.00	29.42
3	2	-0.00	12.40	-0.00	12.66
4	0	0.00	74.48	-0.00	74.46
4	1	-0.00	26.17	-0.00	26.45
4	2	-0.00	12.21	-0.00	12.32
5	0	0.00	75.57	0.01	75.19
5	1	0.01	30.30	0.00	30.73
5	2	-0.00	11.39	-0.00	11.59

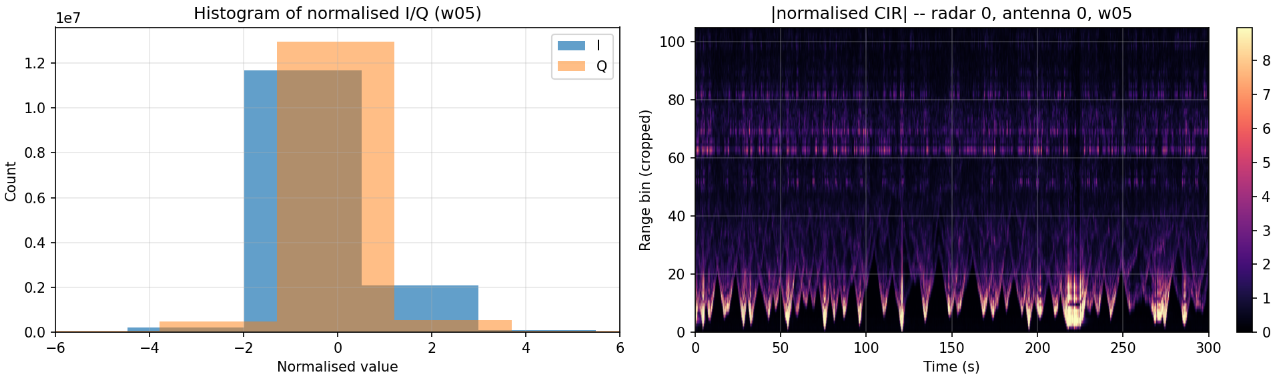
5. Held-out verification

After applying the pipeline to validation and test windows, the per-channel mean should be close to 0 and std close to 1. Significant deviation indicates a domain shift between splits.

Split	mean I	std I	mean Q	std Q
Train (sample)	0.000	0.934	0.000	0.935
Validation	-0.000	0.780	-0.000	0.777
Test	0.000	0.684	-0.000	0.684

6. End-to-end pipeline output

Output of the full pipeline on a representative four-people window (w05). Left: histogram of normalised I/Q values - approximately zero-mean, unit-variance, with the expected heavy tail from moving targets. Right: range-time map of the magnitude of the normalised I/Q on radar 0 antenna 0 - target tracks remain clearly visible after the full pipeline.



7. Resulting model input shape

Quantity	Value
Per-frame input shape	(6, 3, 105, 2)
Per-frame elements	3,780
Per-frame size (float32)	14.77 KB
Reduction vs raw 120 bins	87.5 % of original size